

YONGSHENG MEI

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SUMMARY: Applied Scientist works on the development of Large Language Model (LLM) powering Alexa Plus, significantly improving its agentic capability and enabling its successful launch in 2025. His experience includes **Foundation Models and LLMs, Reinforcement Learning, Learning from Human/AI Feedback**.

SKILL

Programming Language	Python, C, C++, Java, Go, MATLAB, SQL, R
Library/Framework	PyTorch, VERL, NeMo, Scikit-learn, Pandas, TensorFlow, Keras
Tool	Git, Bash, Jupyter, L ^A T _E X, Kubernetes, Docker, Spark, Tableau
Database	MySQL, PostgreSQL, MongoDB, ChromaDB

EDUCATION

The George Washington University <i>PhD in Electrical Engineering</i> Research Areas: Reinforcement Learning, Generative Models, Bayesian Optimization	Sept. 2019 – Oct. 2024 <i>Washington, DC, US</i> GPA: 4.00
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Huazhong University of Science and Technology <i>BE in Automation Engineering</i> Relevant Courses: Pattern Recognition, Control Theory, Computer Architecture	Sept. 2015 – June 2019 <i>Wuhan, Hubei, China</i> GPA: 3.81
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EXPERIENCE

Applied Scientist <i>Amazon, Alexa AI</i>	Nov. 2024 - Present <i>Bellevue, WA, US</i>
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- Developed Alexa’s central foundational LLMs empowering the **public launch of Alexa+**, which is intuitive, intelligent, and responsive, by developing superior LLM solutions and services, increasing the capabilities built into the model and enabling utilizing thousands of APIs and external knowledge sources to provide the best experience for each request across millions of customers and endpoints.
- Owned the LLM post-training with Reinforcement Learning (RL), Learning from Human/AI Feedback (LHF/LAIF), and Evaluation. Leveraging SOTA algorithms such as GRPO and DAPO for training dense/MoE LLM. As validated by Alexa+ more than 100 benchmark metrics, the developed models successfully outperformed external SOTA models (e.g., Claude Sonnet 4.5/4/3.5) on accuracy, latency, and capacity for over 90% of real-user traffic on Alexa+.

Research Assistant <i>The George Washington University</i>	Sept. 2019 - Oct. 2024 <i>Washington, DC, US</i>
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Topic 1: Reinforcement Learning

- Developed an optimized **collective prioritized experience replay** scheme in off-policy multi-agent RL via the prioritization weights assignment. Experiments show that our rewards outperform baseline algorithms by up to 10%. This work has been published at **AAMAS 2023**.
- Applied system-algorithm co-design for RL acceleration, such as using the **data-reuse** strategy in conjunction with experience replay to accelerate the multi-agent RL, where the end-to-end training time reduction is 25.4% (for 32 agents), and exploiting a **virtual cache** to accelerate the sampling phase of the offline RL training, achieving 6× speed-ups.

Topic 2: Multimodal Learning

- Designed a multimodal image segmentation model for brain tumor MRI data. The framework can improve segmentation accuracy via **self-attention** with extracted correlated **common information microstructures** among modalities. The method achieves 92% accuracy for the whole tumor on the BraTS-2020 dataset. This work has been published in Machine Learning for Multimodal Healthcare Data workshop at **ICML 2023**.

Topic 3: Stochastic Modeling and Bayesian Optimization (BO)

- Developed a gradient-aware BO framework to **determine local optimal solutions** in multimodal unknown functions for hyperparameter tuning when using the optimal solution is not physically available.
- Proposed a BO model via Gaussian Cox process on discrete spatiotemporal data to **estimate arrival intensity** (outperforming baselines in 7 out of 9 settings) and **detect regions of interest** based on estimations. This work has been published at **ICLR 2024**.

Topic 4: Network Security

- o Led the development of a reliable **self-synchronizing moving target defense** via customized network and Internet of Things protocols. The system can defend against common attacks, such as MITM and DoS.
- o Introduced a **real-time network intrusion detection** model as a sequence of arriving packets using a causal decision transformer. This work has been accepted at the Artificial Intelligence for Cyber Security workshop at **AAAI 2024**.

Research Engineer Intern

Intelligent Fusion Technology, Inc.

May 2024 – July 2024

Germantown, MD, US

- o Designed and Applied the **transfer learning** algorithm to the **domain adaptation** task to improve the performance of civilian and military vehicle recognition.

Visiting Researcher

Purdue University

June 2023 – Aug. 2023

West Lafayette, IN, US

- o Developed a **continual federated learning** model with time-variant input of each edge device, which leverages the **conditioned diffusion model** that is trained from scratch and fine-tuned from a pre-trained model to generate synthetic data to avoid the catastrophic forgetting problem.

PUBLICATION

1. Egemen Erbayat, **Yongsheng Mei**, Gina Adam, Suresh Subramaniam, Sean Coffey, Nathaniel D Bastian, Tian Lan, *LAMPS: A Learning-based Mobility Planning via Posterior State Inference using Gaussian Cox Process Models*, IEEE INFOCOM Workshop on Deep Learning for Wireless Communications, Sensing, and Security, 2025. [PDF]
2. **Yongsheng Mei**, *Decision-Making via Optimization in Challenging Environment: Bayesian and Multi-Agent Reinforcement Learning Perspectives*, The George Washington University Dissertations, 2025 [PDF]
3. **Yongsheng Mei**, Mahdi Imani, Tian Lan, *Bayesian Optimization through Gaussian Cox Process Models for Spatio-temporal Data*, International Conference on Learning Representations (ICLR), 2024. [PDF]
4. **Yongsheng Mei**, Hanhan Zhou, Tian Lan, *Projection-Optimal Monotonic Value Function Factorization in Multi-Agent Reinforcement Learning*, International Conference on Autonomous Agents and Multiagent Systems (AAMAS), 2024. [PDF]
5. Yurong Chen, **Yongsheng Mei**, Tian Lan, Guru Venkataramani, *Exploring Effective Fuzzing Strategies to Analyze Communication Protocols*, ACM Digital Threats: Research and Practice, 2024. [PDF]
6. Jingdi Chen, Hanhan Zhou, **Yongsheng Mei**, Gina Adam, Nathaniel Bastian, Tian Lan. *Real-time Network Intrusion Detection via Decision Transformers*, AAAI workshop on Artificial Intelligence for Cyber Security (AICS), 2024. [PDF]
7. Jiayu Chen, Bhargav Ganguly, Yang Xu, **Yongsheng Mei**, Tian Lan, Vaneet Aggarwal, *Deep Generative Models for Offline Policy Learning: Tutorial, Survey, and Perspectives on Future Directions*, Transactions on Machine Learning Research, 2024 [PDF]
8. **Yongsheng Mei**, Tian Lan, Mahdi Imani, Suresh Subramaniam, *A Bayesian Optimization Framework for Finding Local Optima in Expensive Multi-Modal Functions*, European Conference on Artificial Intelligence (ECAI), 2023. [PDF]
9. **Yongsheng Mei**, Tian Lan, Guru Venkataramani, *Exploiting Partial Common Information Microstructure for Multi-Modal Brain Tumor Segmentation*, ICML workshop on Machine Learning for Multimodal Healthcare Data, 2023. [PDF]
10. Kailash Gogineni, **Yongsheng Mei**, Peng Wei, Tian Lan, Guru Venkataramani, *AccMER: Accelerating Multi-agent Experience Replay with Cache Locality-aware Prioritization*, IEEE International Conference on Application-specific Systems, Architectures and Processors (ASAP), 2023. [PDF]
11. **Yongsheng Mei**, Hanhan Zhou, Tian Lan, Guru Venkataramani, Peng Wei, *MAC-PO: Multi-Agent Experience Replay via Collective Priority Optimization*, International Conference on Autonomous Agents and Multiagent Systems (AAMAS), 2023. [PDF]
12. Kailash Gogineni, **Yongsheng Mei**, Guru Venkataramani, Tian Lan, *Verify-Pro: A Framework for Server Authentication using Communication Protocol Dialects*, IEEE Military Communications Conference (MILCOM), 2022. [PDF]
13. **Yongsheng Mei**, Kailash Gogineni, Tian Lan, Guru Venkataramani, *MPD: Moving Target Defense through Communication Protocol Dialects*, International Conference on Security and Privacy in Communication Networks (SecureComm), 2021. [PDF]
14. Hongfa Xue, **Yongsheng Mei**, Kailash Gogineni, Guru Venkataramani, Tian Lan, *Twin-Finder: Integrated Reasoning Engine for Pointer-related Code Clone Detection*, International Workshop on Software Clones, 2020. [PDF]

AWARD

2025 GW ECE Best Dissertation Honorable Mention Award

2023 European Conference on Artificial Intelligence *Call to Arms* Award

2020 – 2023 GW University Fellowship

2019 HUST Outstanding Graduates